

Summary Data on Acid Cropland and Lime Requirements in Rwanda

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Last Updated: November 4, 2024

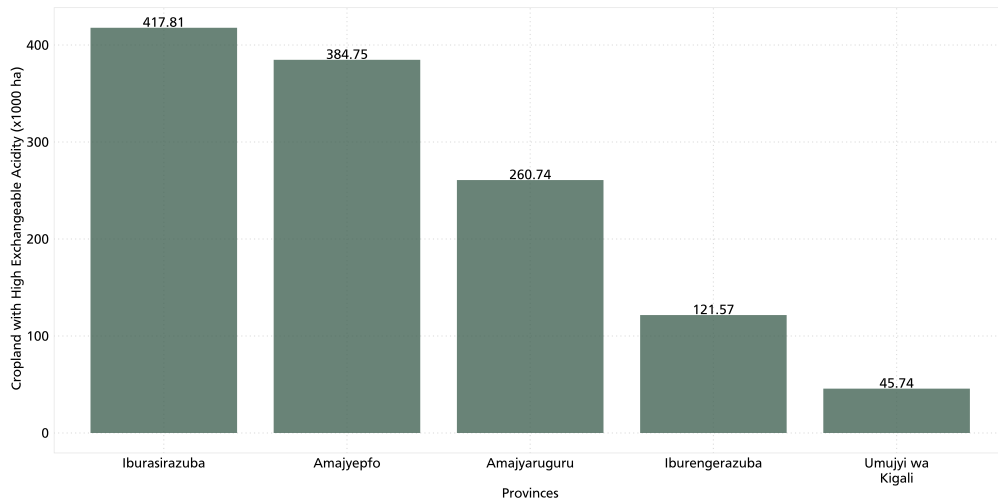
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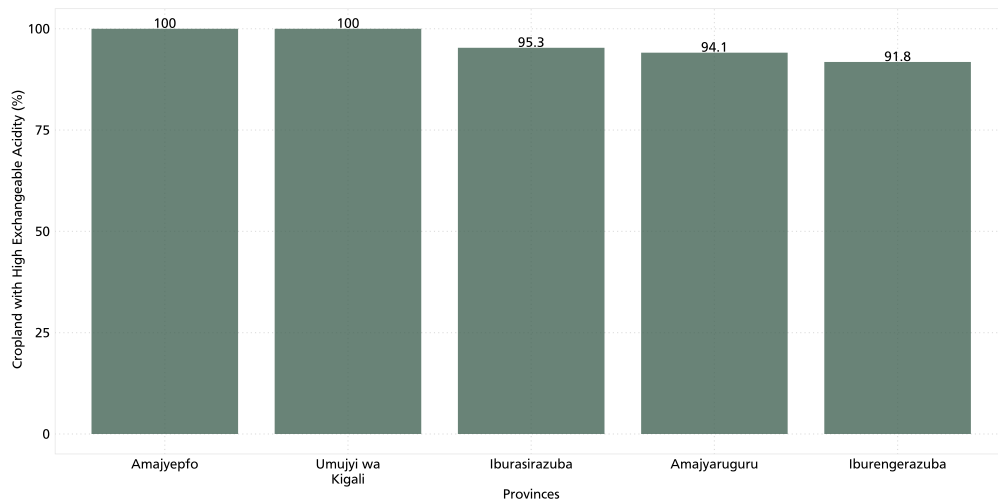
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Rwanda**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

Top 10 Provinces in Rwanda with the highest acidic cropland



Top 10 Provinces in Rwanda with the highest acidic cropland (Percentage of total cropland)



The chart is a stacked bar graph with the y-axis labeled 'Crop Area (x1000 ha)' ranging from 0 to 400. The x-axis is labeled 'Provinces' and lists five provinces: Iburasirazuba, Amajepfo, Amajyaruguru, Iburengerazuba, and Umuji wa Kigali. The legend on the right identifies eight crop types: Bean (dark green), Cassava (yellow), Maize (light blue), Other-Cereal (brown), Other-Legume (red), Other-Non-food (teal), Other-RTBs (orange), and Potato (dark orange). Iburasirazuba has the highest total crop area, exceeding 400,000 ha, with significant portions in Cassava, Other-Legume, and Other-RTBs. Amajepfo follows with a total area around 380,000 ha, dominated by Bean and Other-Cereal. Amajyaruguru has a total area of approximately 280,000 ha, with Maize and Bean being major components. Iburengerazuba and Umuji wa Kigali have much smaller total crop areas, around 130,000 ha and 40,000 ha respectively, with a more diverse mix of crops.

Province	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Iburasirazuba	40	140	20	20	70	40	90	10
Amajepfo	120	20	20	110	5	5	90	10
Amajyaruguru	120	10	70	30	5	15	30	10
Iburengerazuba	20	10	60	15	5	5	20	10
Umuji wa Kigali	30	5	5	5	2	2	5	2

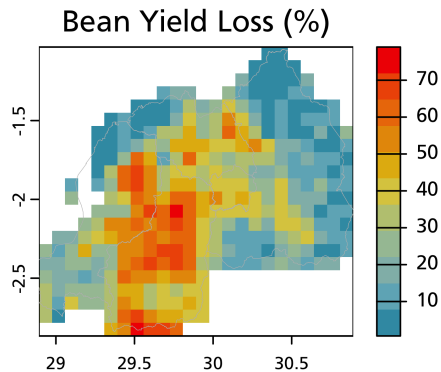
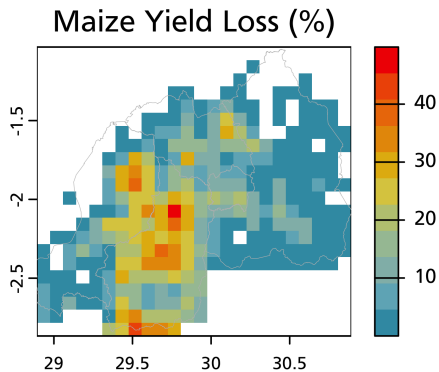
└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland

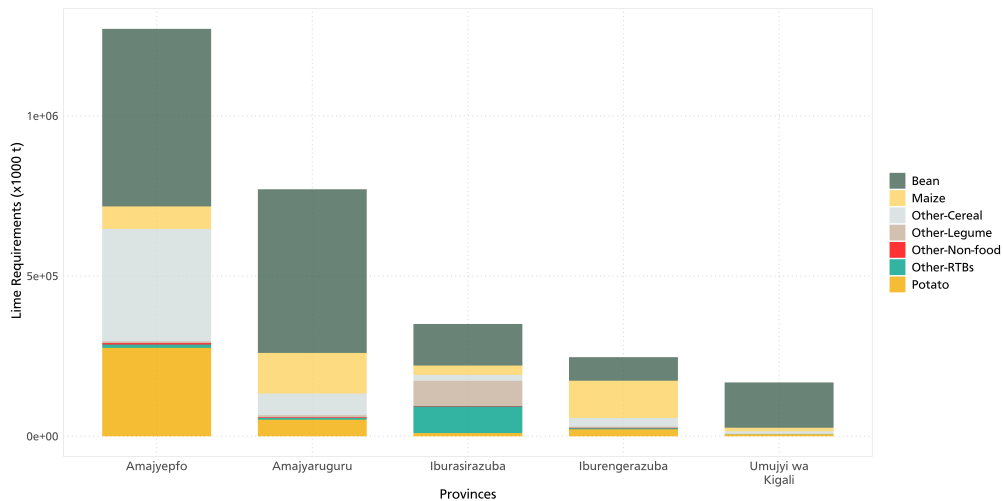
Total crop area in the top-10 most acidic Provinces, not segregated by acidic cropland (table)

Provinces	Bean	Cassava	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	124.66	5.03	72.61	29.67	2.67	11.32	3.68	27.53
Amajyepfo	125.33	16.69	25.34	115.82	1.59	4.70	3.50	91.79
Iburasirazuba	37.96	143.45	19.67	22.34	71.59	36.79	97.00	9.47
Iburengerazuba	18.36	9.56	62.46	15.18	2.88	3.87	4.47	15.66
Umujyi wa Kigali	31.90	2.93	4.72	3.03	0.36	0.26	0.16	2.39

Yield loss due to soil acidity for Maize and Bean in Rwanda



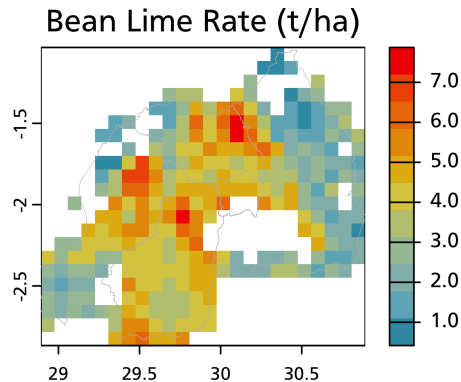
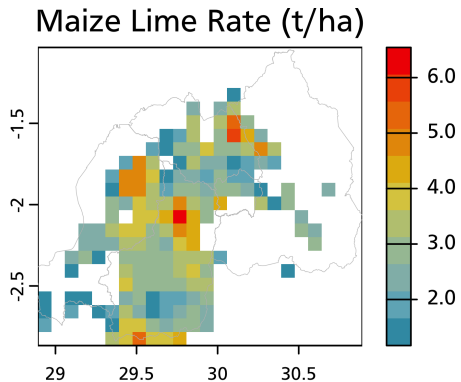
Lime requirements (t) for acid soil remediation in Rwanda



Lime requirements (1000 t) for acid soil remediation in Rwanda (table)

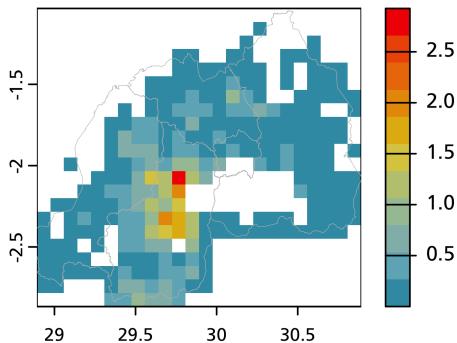
Provinces	Bean	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Amajyaruguru	510.71	126.90	67.61	6.31	1.92	5.83	51.78
Amajyepfo	554.12	70.34	350.70	5.48	4.64	10.48	276.00
Iburasirazuba	129.56	29.07	18.75	78.50	1.56	82.96	10.16
Iburengerazuba	73.38	116.54	26.91	3.81	1.22	3.23	21.92
Umujyi wa Kigali	141.43	10.92	7.82	1.45	0.03	0.42	6.08

Lime rate for Maize and Bean in Rwanda

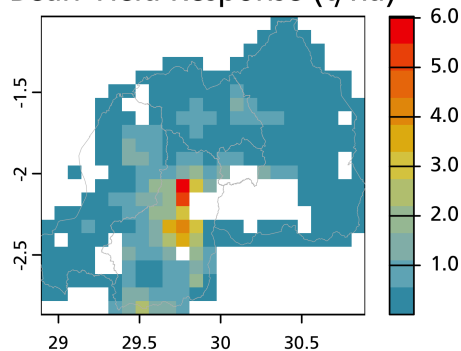


Yield response to lime application in Rwanda

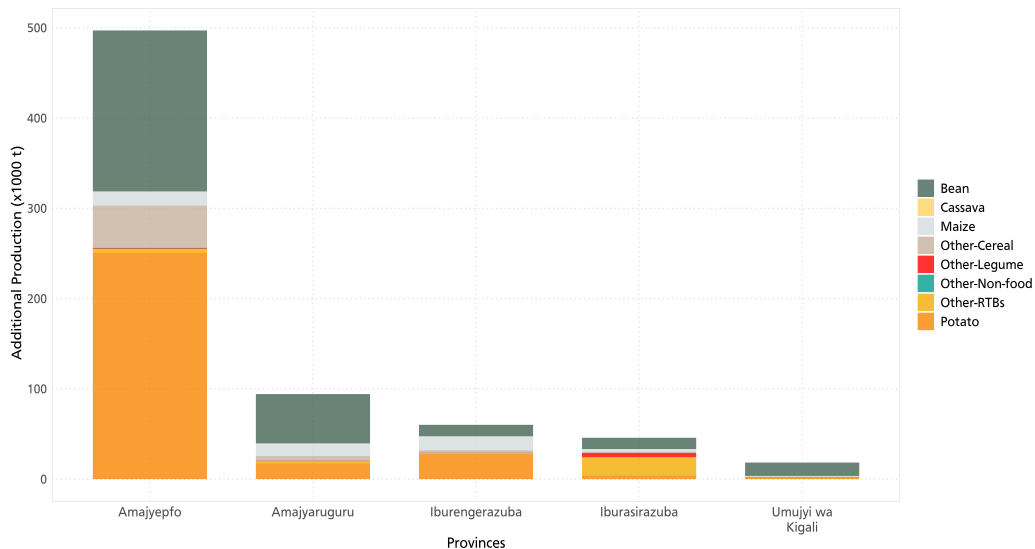
Maize Yield Response (t/ha)



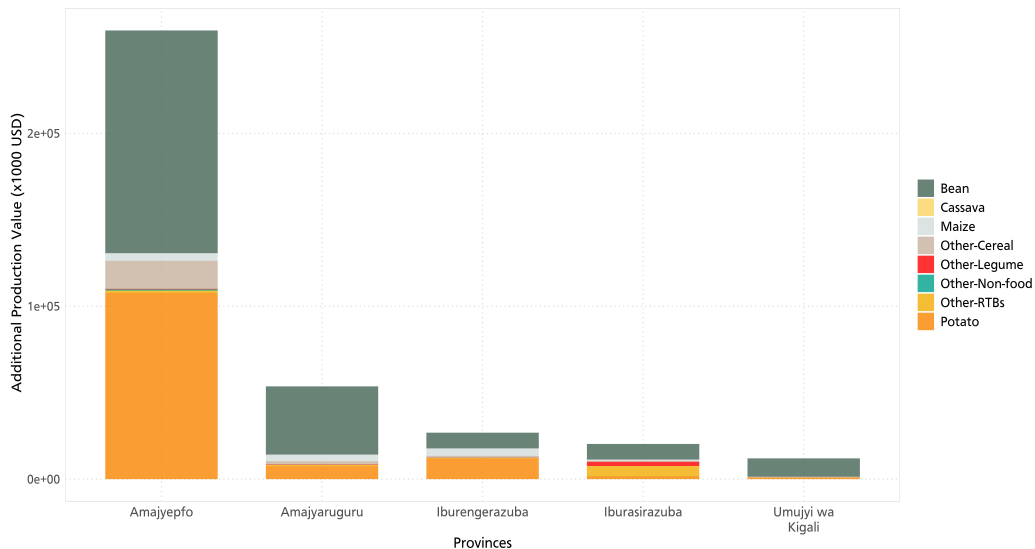
Bean Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Provinces in Rwanda

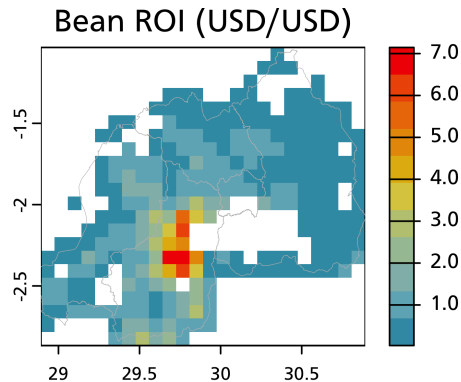
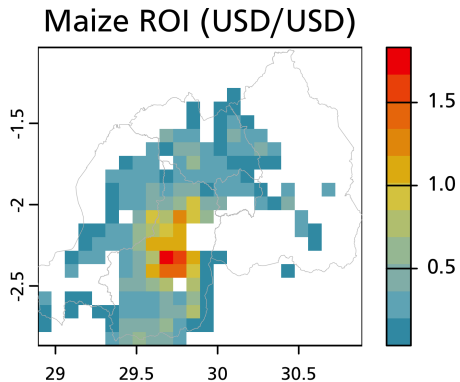


Potential for additional production value (US\$) in top 10 acidic Provinces in Rwanda



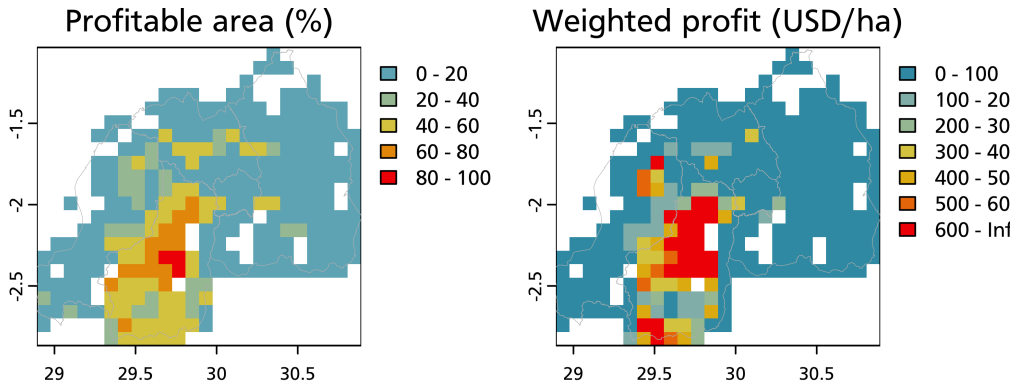
- ROI for Lime Application
- Return on investment for lime application in Rwanda

Return on investment for lime application in Rwanda



- └ Profitability of Lime Application
- └ Profitability of lime application in Rwanda

Profitability of lime application in Rwanda



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.